Project-05

Bachelor of Computer Application – Data science
Semester – V

Sub: Deep Learning

Topic: Activation Functions Using Python

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Activation Function Using Python

```
import numpy as np
import matplotlib.pyplot as plt

#Task 1 : define activation function

#sigmoid functions def
sigmoid(x):   return 1/(1+
np.exp(-x))


#relu function def
relu(x):   return
np.maximum(0,x)


# Tanh Function def
tanh(x):   return
np.tanh(x)

#Task 2



#input values
x_value = np.array([-5,-3,-1,0,1,3,5])


#Apply activation finctions
sigmoid_output = sigmoid(x_value)
relu_output = relu(x_value)
tanh_output = tanh(x_value)


#display outputs
print("input Values")
print(x_value)

print("\nSigmoid Output")
print(sigmoid_output)
input Values

[-5 -3 -1  0  1  3  5]
Sigmoid Output
[0.00669285 0.04742587 0.26894142 0.5        0.73105858 0.95257413
 0.99330715]

#Task 3
#generated continous input values for plotting
```

```
x= np.linspace(-5,5,100)


#calculate outputs y_sigmoid = sigmoid(x) y_relu = relu(x) y_tanh =
tanh(x)

#plot all  activation functions plt.figure(figsize=(10,6))
plt.plot(x,y_sigmoid,label='sigmoid') plt.plot(x,y_relu,label='relu') plt.plot(x,y


plt.title("Comparison Of Activation Finctions") plt.xlabel("input value") plt.ylab
Value")

plt.grid(True) plt.legend() plt.show()
```